Experiment 11: Hive Operations - USE, ALTER, DROP (Windows)

Aim

To download, install, and configure Apache Hive on a Windows operating system and implement

schema modification and management operations using the USE, ALTER, and DROP commands. Procedure
1. Extract the downloaded Apache Hive distribution package files directly into a clean root disk path

destination folder such as C:\hive. 2. Configure the system environment variables panel by assigning HIVE_HOME to C:\hive and adding
the binary folder route directory path %HIVE_HOME%\bin into your system path. 3. Boot up your underlying core Apache Hadoop backend architecture layers by running start-
dfs.cmd and start-yarn.cmd inside an administrative Command Prompt terminal window. 4. Set up the required underlying HDFS warehouse paths (/tmp and /user/hive/warehouse) with
expanded read and write access clearance permissions. 5. Initialize the relational database metastore framework layout on your local machine by executing
the schema initialization command line tool schematool -dbType derby -initSchema. 6. Open the interactive Apache Hive client shell by executing the hive runtime driver command script
directly from the command window interface. 7. Execute DDL queries to create a database container, switch database workspaces using the USE

statement, modify metadata properties with the ALTER statement, and remove objects using the

DROP statement. Commands

Windows Console Pre-requisite Commands
cmd
:: 1. Confirm that NameNode, DataNode, and Derby metastores are ready for connections jps

:: 2. Open the active interactive Apache Hive execution client environment terminal shell
hive

Use code with caution. Interactive Hive Shell Queries

sql -- 3. Create a unique isolated database workspace container mapping environmentCREATE

DATABASE IF NOT EXISTS inventory_db;
- - 4. Switch the current active shell connection context to target the new workspace

USE inventory_db;
- - 5. Create a basic placeholder relational table structure inside the current
schemaCREATE TABLE items_backup (item_id INT, item_name STRING);
- - 6. Alter the schema of the table to inject extra tracking column attributesALTER
TABLE items_backup ADD COLUMNS (unit_price DOUBLE, stock_count INT);
- - 7. Permanently delete the table schema layout and its data files from your
clusterDROP TABLE items_backup;

Use code with caution. Execution & Output

text
C:\hive\bin>hive

hive> CREATE DATABASE IF NOT EXISTS inventory_db;
OK
Time taken: 0.38 seconds

hive> USE inventory_db;
OK
Time taken: 0.03 seconds

hive> CREATE TABLE items_backup (item_id INT, item_name STRING);
OK
Time taken: 0.29 seconds

hive> ALTER TABLE items_backup ADD COLUMNS (unit_price DOUBLE, stock_count INT); OK
Time taken: 0.52 seconds

hive> DESCRIBE items_backup;
OK
col_name data_type comment
item_id int
item_name string

unit_price double
stock_count int
Time taken: 0.15 seconds

hive> DROP TABLE items_backup;
OK
Time taken: 0.41 seconds